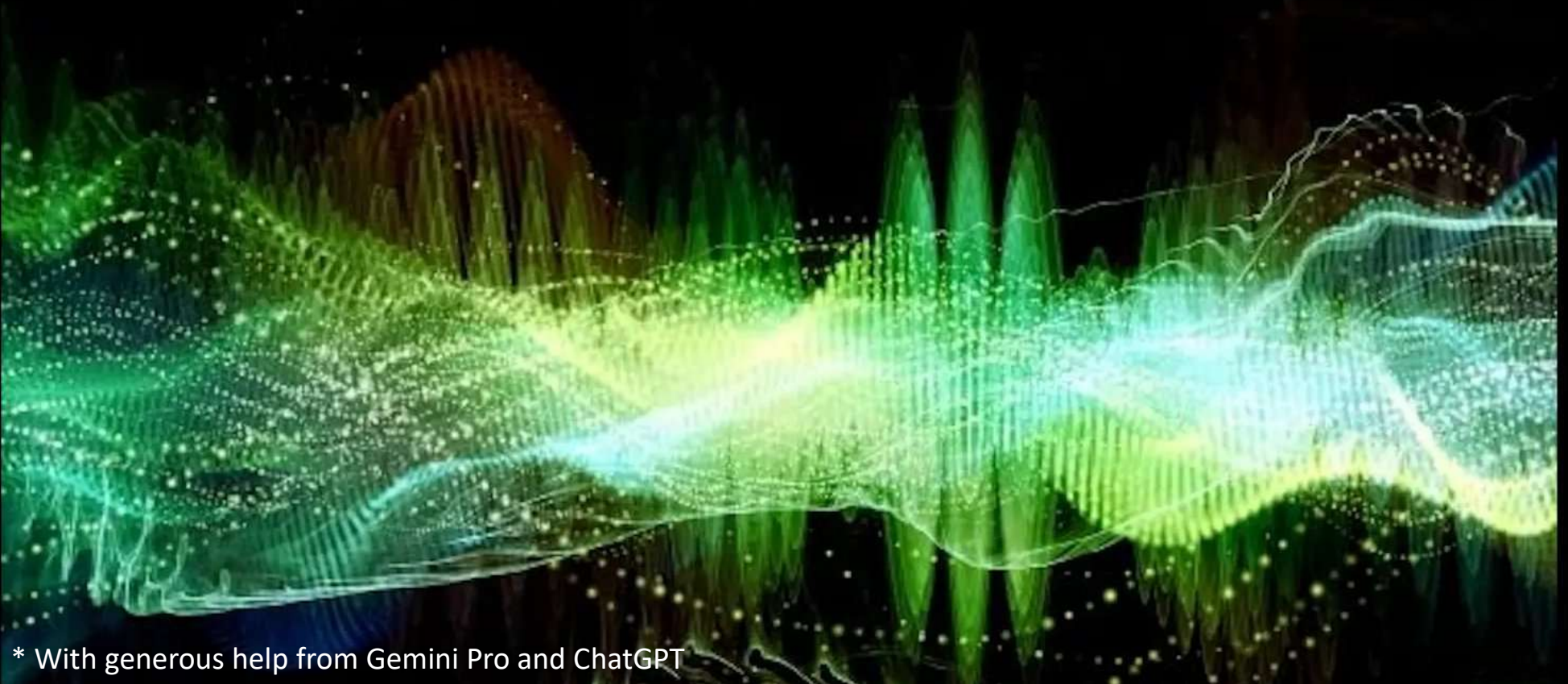


ECEN 4/5632 Theory and Application of Digital Filters

Lecture 8, 9-17-26

Peter Mathys*



* With generous help from Gemini Pro and ChatGPT

Midterm Exam 1

Thursday September 24, 9:30 – 10:45 am, ECCR 131 for in-class students.

Between Sep. 24 and Sep. 30 for on-line students who must supply email address of exam proctor.

Covers Lectures 1 through 8 and Problem Sets 1 through 4

Exam problems will be similar to Problem Set questions

Closed book, notes, no cheat sheets. Reference formulas will be printed on exam.

No laptops, tablets, phones, basic scientific calculator o.k.

IIR Design, Bilinear Transformation, & Data Preprocessing

From analog prototypes to stable digital filters—and the phase choices hidden in ML preprocessing



Where this 75-minute lecture is going

0–15 min • Analog prototypes

Butterworth, Chebyshev I/II, and elliptic tradeoffs.

15–40 min • Bilinear transform

Map the analog LHP into the digital unit disk while preserving stability.

40–60 min • Warping + prewarp

Derive $\Omega(\omega)$ and hit the desired digital critical frequencies.

60–75 min • ML preprocessing

Causal IIR versus forward-backward zero-phase filtering and deployment implications.

DSP bridge: IIR design gives efficient recursive filters; ML preprocessing must also account for phase, causality, and train/deployment consistency.

Classical DSP supplies structure and constraints; machine learning either reuses that structure or learns parameters inside it.

Why accept feedback and nonlinear phase?

Efficiency

Sharp magnitude selectivity can often be obtained with far fewer coefficients than an FIR meeting comparable magnitude specifications.

Tradeoff

The recursive denominator introduces poles and typically nonlinear phase / frequency-dependent group delay.

ML preprocessing

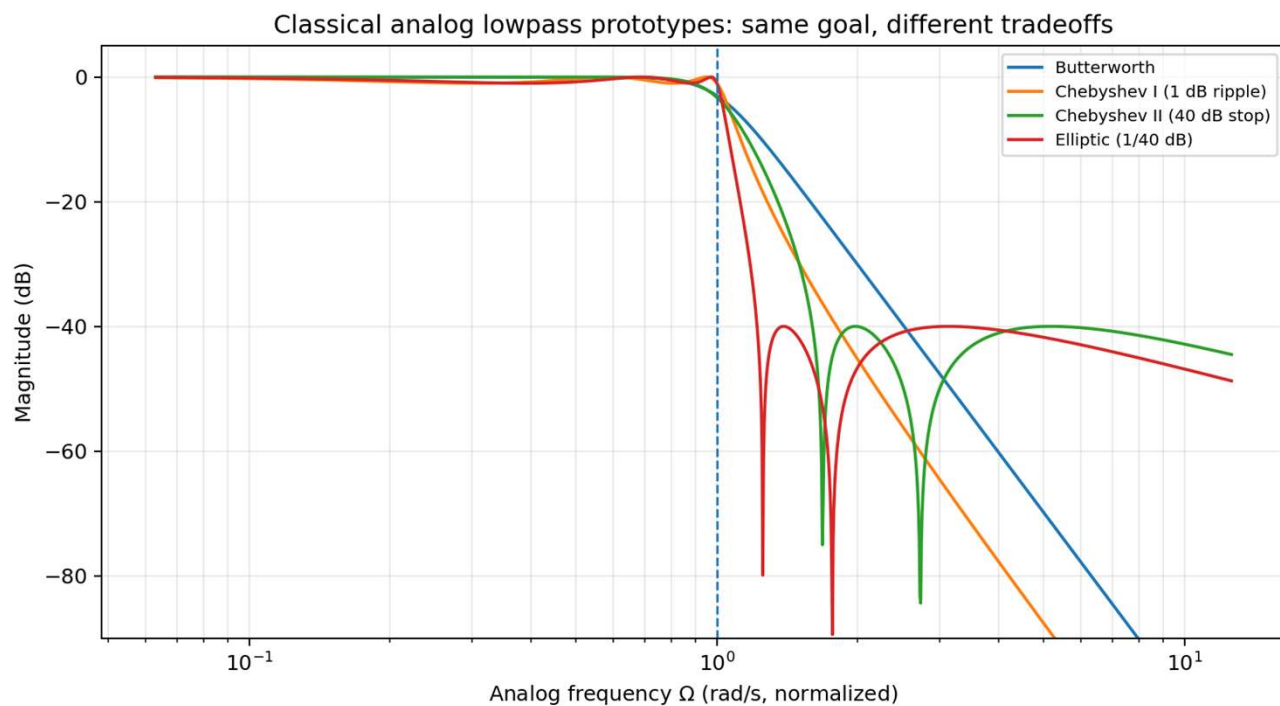
For offline denoising or smoothing, compact IIR filters are attractive—but preprocessing must match the temporal assumptions of deployment.

$$H(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{1 + \sum_{k=1}^N a_k z^{-k}}$$

FIR zeros
z-plane

IIR poles
z-plane

Four classical ways to spend ripple and transition width



Butterworth

Maximally flat magnitude in the passband; monotonic.

Chebyshev

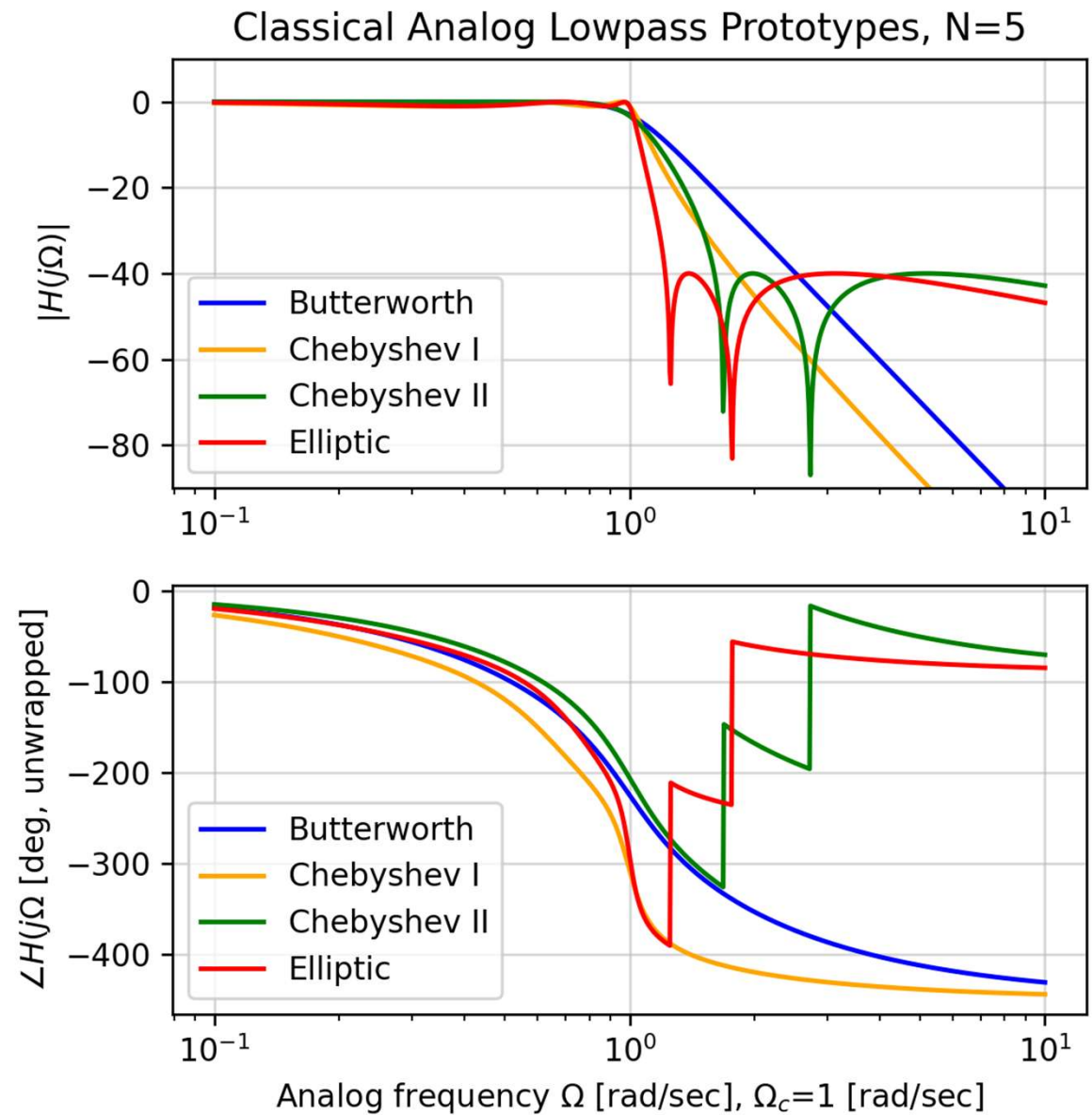
Spend ripple in one band to sharpen the transition.

Elliptic

Cover

Ripple in both bands; steepest transition for a given order/specification.

Magnitude and Phase of Analog Lowpass Prototypes



$$s = j\Omega \Rightarrow \Omega = \frac{s}{j}$$

Maximally flat magnitude response

$$= H_c(j\Omega) \cdot H_c(-j\Omega) \Rightarrow H_c(s) \cdot H_c(-s) = \frac{1}{1 + \left(\frac{s}{j\Omega_c}\right)^{2N}}$$

$$|H_c(j\Omega)|^2 = \frac{1}{1 + \left(\frac{\Omega}{\Omega_c}\right)^{2N}}$$

find poles in s-plane from this

$$|H_c(j\Omega_c)| = \frac{1}{\sqrt{2}} \Rightarrow -3.01 \text{ dB}$$

$$H(j\Omega) = \int h(t) e^{-j\Omega t} dt$$

$$H^*(j\Omega) = \int h^*(t) e^{j\Omega t} dt = \int h(t) e^{j\Omega t} dt = H(-j\Omega)$$

↑
if $h(t)$ is real

$$|H(j\Omega)|^2 = H(j\Omega) \cdot H(-j\Omega)$$

Butterworth

$$\frac{1}{1 + \left(\frac{s}{j\Omega_c}\right)^{2N}}$$

find poles of this function

Order N

Higher N steepens the transition around Ω_c .

No passband ripple

Butterworth is attractive when smooth magnitude behavior matters more than the narrowest possible transition.

Good answer

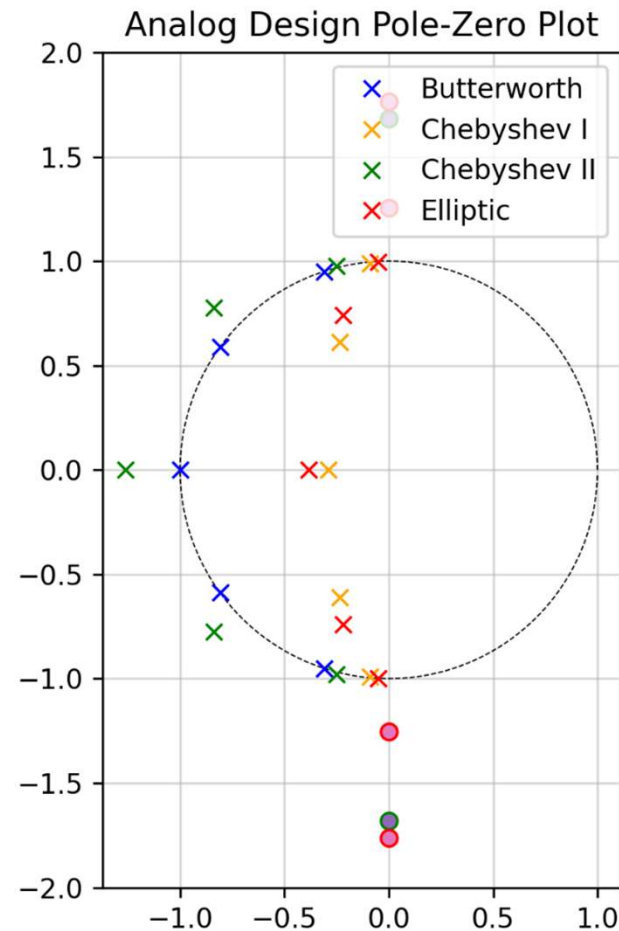
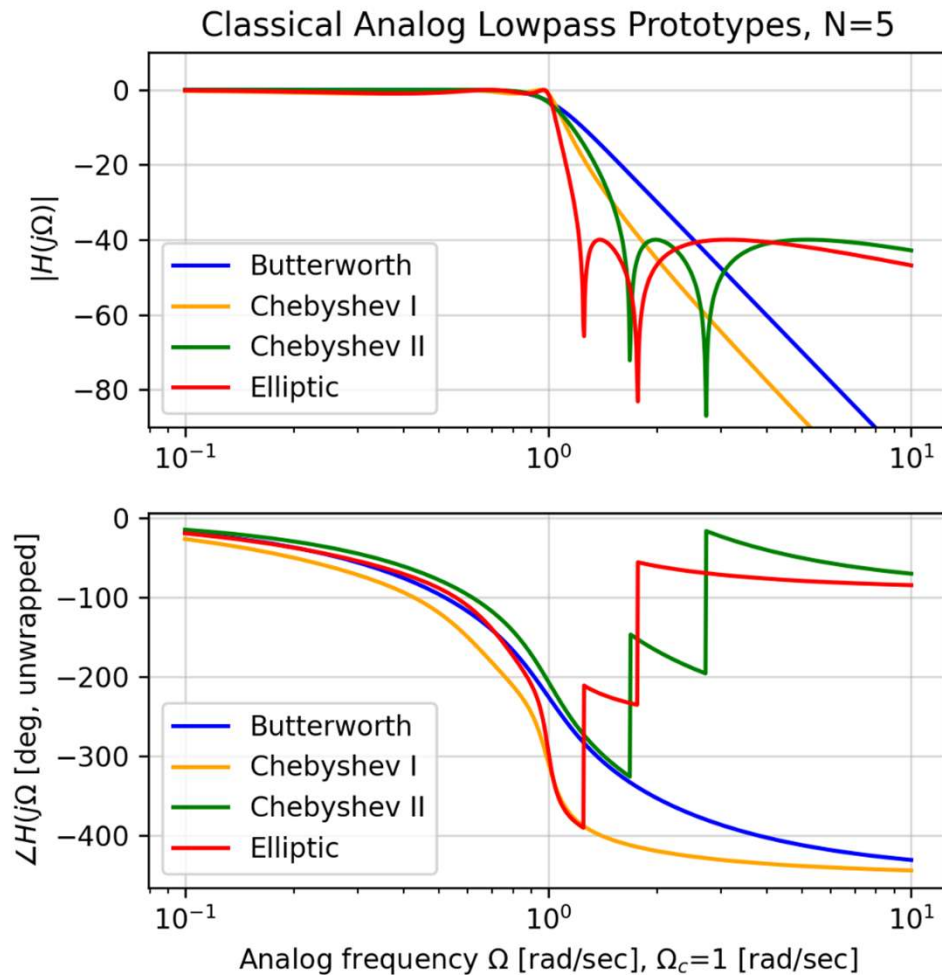
At the cutoff, the squared magnitude is 1/2 regardless of N.

pick poles in left half of s-plane

↓
obtain poles

of B-Butterworth filter

Magnitude, Phase, and Pole-Zero (in s-Plane) Plots for Analog Designs



What increasing order buys—and what it costs

$$\Omega \gg \Omega_c : |H_c(j\Omega)| \approx \left(\frac{\Omega_c}{\Omega}\right)^N$$

$$|H_c(j\Omega)|^2 = \frac{1}{1 + \left(\frac{\Omega}{\Omega_c}\right)^{2N}}$$

Magnitude slope

Each additional pole increases asymptotic attenuation by about 20 dB/decade.

Implementation cost

Higher order means more states / sections and greater numerical sensitivity in direct-form implementations.

Practice

Use second-order sections for moderate/high-order digital IIR implementations when possible.

if $\Omega \gg \Omega_c$ (stopband) $|H_c(j\Omega)| \approx \left(\frac{\Omega_c}{\Omega}\right)^N$

$$20 \log_{10} \left(\frac{\Omega_c}{\Omega}\right)^N = 20N \log_{10}(\Omega_c) - 20N \log_{10}(\Omega)$$

fixed

goes down by -20dB/decade per N

Where the ripple is allowed to live

Prototype	Passband	Stopband	Transition / comment
Butterworth	monotonic	monotonic	maximally flat; smooth
Chebyshev I	<u>equiripple</u>	monotonic	sharper for allowed passband ripple
Chebyshev II	monotonic	<u>equiripple</u>	spend ripple in stopband
Elliptic	<u>equiripple</u>	<u>equiripple</u>	steepest transition for given order/spec

Engineering habit

Choose the prototype from passband/stopband tolerances and order requirements—not from a universal ranking.

CONCEPT CHECK

Which prototype sounds plausible?

A • Sensor calibration

Passband amplitude must be smooth;
moderate transition width is acceptable.

Butterworth

B • Tight transition

A small amount of controlled passband
ripple is acceptable to lower order.

Chebyshev I

C • Most aggressive

Ripple is acceptable in both bands and
minimum order is important.

Elliptic

CONCEPT CHECK

Which prototype sounds plausible?

A • Sensor calibration

Passband amplitude must be smooth;
moderate transition width is acceptable.

B • Tight transition

A small amount of controlled passband
ripple is acceptable to lower order.

C • Most aggressive

Ripple is acceptable in both bands and
minimum order is important.

Good answer

Butterworth is a natural first choice
because it is monotonic and maximally flat.

Good answer

Chebyshev I trades passband ripple for a
sharper transition.

Good answer

Elliptic is often the most aggressive
classical choice.

What must a useful domain map preserve?

Stability

Analog stable poles satisfy $\text{Re}\{s\} < 0$. They must map to $|z| < 1$.

Frequency axis

The analog $j\Omega$ axis should map to the digital unit circle $z = e^{j\omega}$.

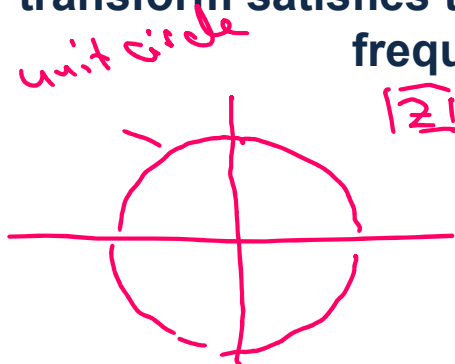
One-to-one mapping

Avoid aliasing of multiple analog frequencies onto the same digital frequency.

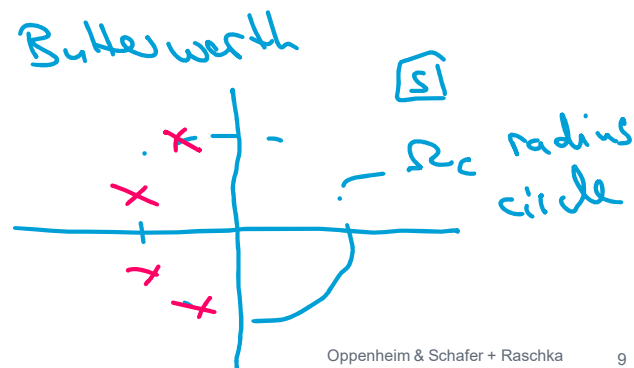
need map from s -plane to z -plane s.t. left half of s -plane maps into inside of unit circle in z -plane (for causal, stable filters)

The bilinear transform satisfies these properties—but its price is nonlinear frequency warping.

different from Ω_c circle!



how to get to



The substitution that maps the half-plane to the disk

$$s = \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}}$$

$$z = \frac{1 + sT/2}{1 - sT/2}$$

Interpret T

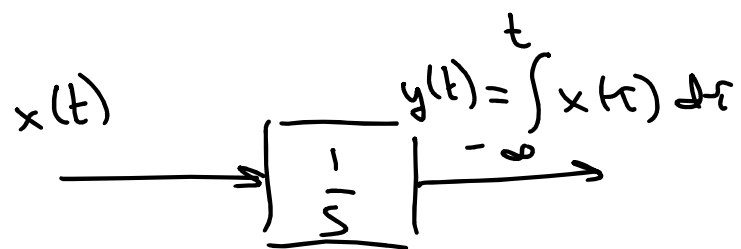
T is the sampling period used in the analog-to-digital mapping.

Important

This is not the impulse-invariance map $z = e^{sT}$. The bilinear transform avoids frequency aliasing but warps the axis.

$$s = \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}}$$

Solve for z : $z = \frac{1 + \frac{T}{2}s}{1 - \frac{T}{2}s}$



CT integrator

we want to replace it with
a DT approximation

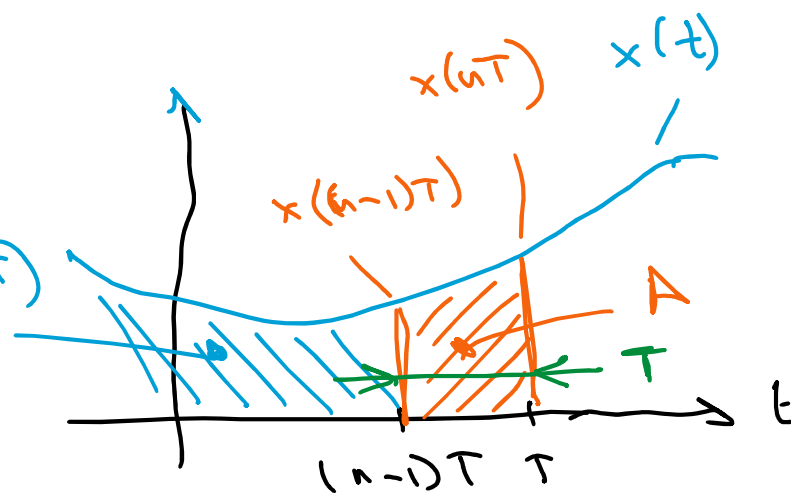
$$y(nT) = \int_{-\infty}^{nT} x(\tau) d\tau = \underbrace{\int_{-\infty}^{(n-1)T} x(\tau) d\tau}_{= y((n-1)T)} + \underbrace{\int_{(n-1)T}^{nT} x(\tau) d\tau}_{= A} = y((n-1)T) + A$$

$$A \approx \frac{x((n-1)T) + x(nT)}{2} \cdot T$$

$$y(nT) \rightarrow y[n]$$

$$y[n] = y[n-1] + \frac{T}{2} (x[n] + x[n-1])$$

↑ recursion for $y[n]$



$$y[n] = y[n-1] + \underbrace{\frac{T}{2}(x[n] + x[n-1])}_{=A}$$

in z -domain

$$(1 - z^{-1})Y(z) = \frac{T}{2}(1 + z^{-1})X(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{T}{2} \frac{1 + z^{-1}}{1 - z^{-1}} = \frac{1}{s}$$

we made approx
of CT integrator

$$\Rightarrow s = \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}}$$

$$\text{solve for } z : z = \frac{1 + \frac{T}{2}s}{1 - \frac{T}{2}s}$$

Why the analog left-half plane lands inside the unit circle

$$|z|^2 = \frac{|1 + sT/2|^2}{|1 - sT/2|^2} \quad \leftarrow 1 - sT/2 \text{ denominator}$$

$$s = \sigma + j\Omega, \sigma < 0 \Rightarrow |1 + sT/2| < |1 -$$

$$\therefore |z| < 1$$

left half of
s-plane
 $\sigma < 0$

$$< |1 - sT/2|$$



$$|z|^2 < 1$$

$$\Rightarrow |z| < 1 \quad \text{inside of unit circle}$$

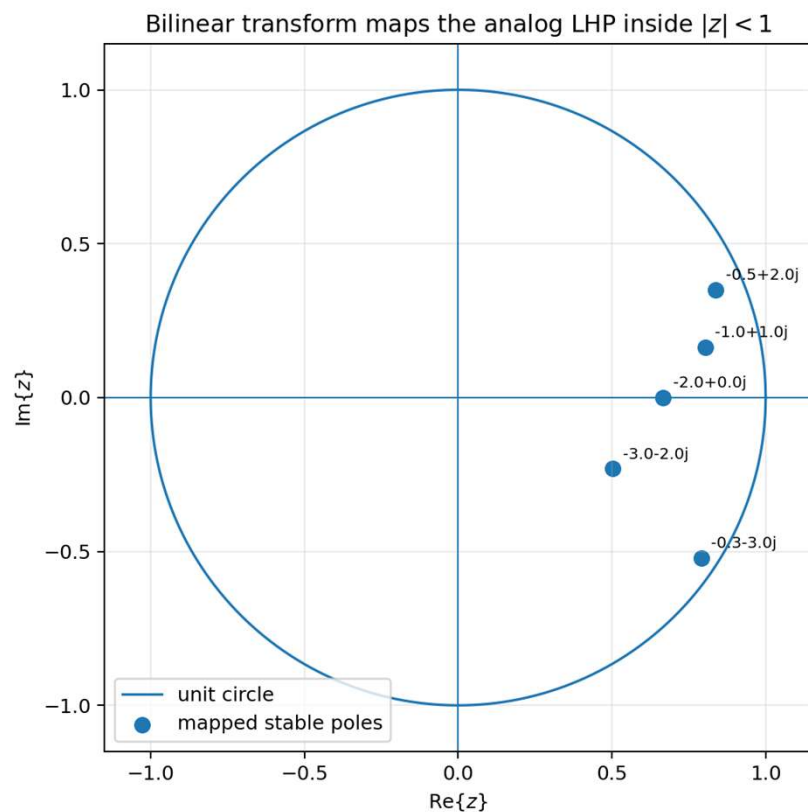
Good answer

The negative real part makes the numerator magnitude smaller than the denominator magnitude, so a stable analog pole maps inside the digital unit circle.

Boundary

For $\sigma=0$, numerator and denominator magnitudes are equal, hence $|z|=1$.

Representative analog stable poles map inside $|z|=1$



Interpretation

Every shown analog pole has negative real part. After the bilinear map, every digital pole lies strictly inside the unit circle.

This is the guarantee we want

Start from a stable analog prototype and the mapped digital filter remains stable, barring numerical implementation issues.

The analog $j\Omega$ axis maps exactly onto the unit circle

$$z = \frac{1 + \frac{T}{2}s}{1 - \frac{T}{2}s} \quad z = e^{j\omega}$$

$$s = j\Omega, \quad z = e^{j\omega}$$

$$j\Omega = \frac{2}{T} \frac{1 - e^{-j\omega}}{1 + e^{-j\omega}}$$

$$\frac{1 - e^{-j\omega}}{1 + e^{-j\omega}} = j \tan\left(\frac{\omega}{2}\right)$$

$$\angle z = \tan^{-1}\left(\frac{\Omega T}{2}\right) - \tan^{-1}\left(-\frac{\Omega T}{2}\right) = 2 \tan^{-1}\left(\frac{\Omega T}{2}\right) = \omega$$

Endpoints

$\Omega=0$ maps to $\omega=0$. As $|\Omega| \rightarrow \infty$, the digital frequency approaches $\pm\pi$ and z approaches -1 .

No aliasing

The entire analog frequency axis maps one-to-one into $-\pi < \omega < \pi$.

set $\sigma=0$ $s = j\Omega$

$$z = \frac{1 + j\frac{\Omega T}{2}}{1 - j\frac{\Omega T}{2}}$$

$$\Rightarrow |z|^2 = \frac{1 + \left(\frac{\Omega T}{2}\right)^2}{1 + \left(\frac{\Omega T}{2}\right)^2} = 1 \quad |z| = 1$$

set $z = e^{j\omega}$

sitting on unit circle

The price of the one-to-one map

$$\Omega = \frac{2}{T} \tan\left(\frac{\omega}{2}\right)$$

$$\omega = 2 \tan^{-1}\left(\frac{\Omega T}{2}\right)$$

$$2 \tan^{-1}\left(\frac{\Omega T}{2}\right) = \omega$$

$$\frac{\omega}{2} = \tan^{-1}\left(\frac{\Omega T}{2}\right) \Rightarrow$$

$$\Omega = \frac{2}{T} \tan\left(\frac{\omega}{2}\right)$$

in design process start from ω_c → find Ω_c using

Low frequency

For small ω , $\tan(\omega/2) \approx \omega/2$, so $\Omega \approx \omega/T$.

Near Nyquist

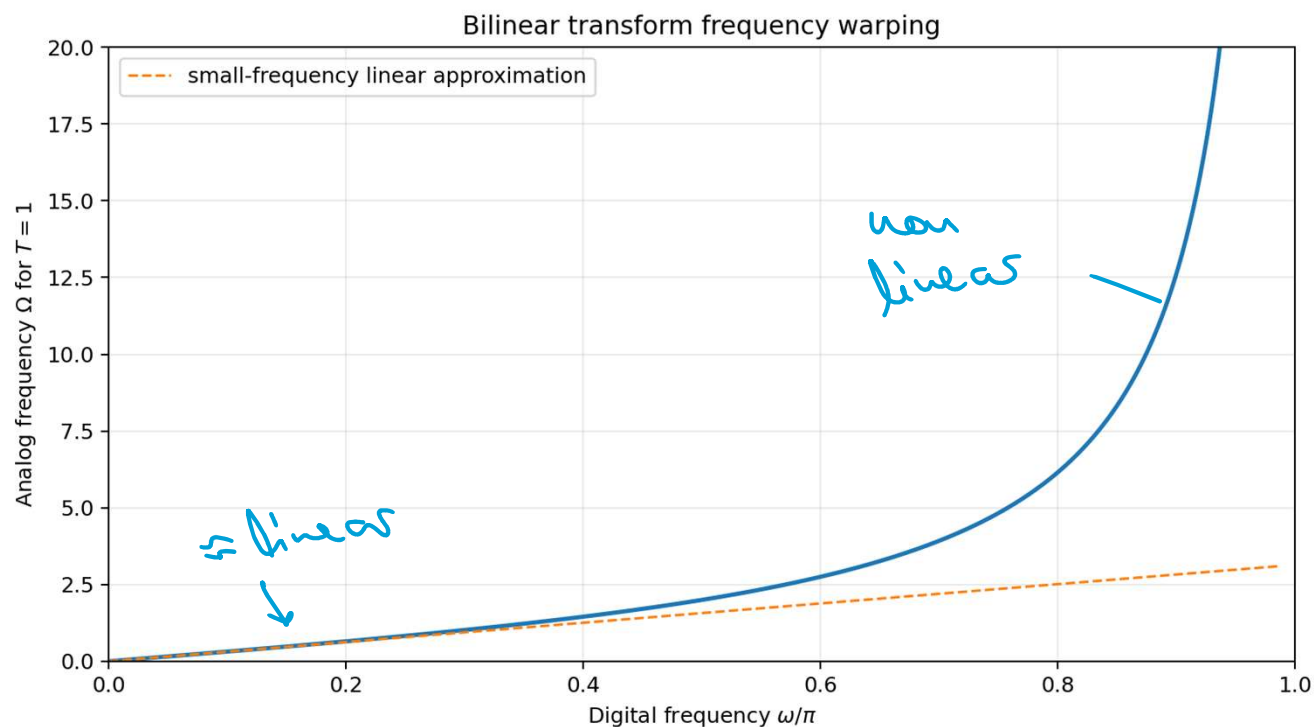
The map becomes strongly nonlinear; equal analog frequency intervals do not remain equal digital intervals.

with
bilinear
transf

map from s to z
domain

design analog
prototype

Near Nyquist, the analog axis is stretched dramatically

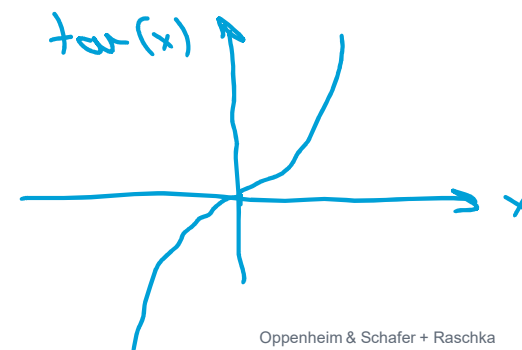


Read the curve

The relationship is nearly linear near DC but diverges as $\omega \rightarrow \pi$.

Design consequence

If you design the analog cutoff at the unwrapped numeric value, the final digital cutoff misses its target.



PREWARPING

Design the analog prototype at the frequency that maps to the desired digital edge

$$\Omega_k = \frac{2}{T} \tan \left(\frac{\omega_k}{2} \right)$$

Design sequence

1 • Digital spec

ω_p , ω_s , ripple / attenuation

2 • Prewarp

map each critical digital frequency to Ω

3 • Analog design

design Butterworth / Chebyshev / elliptic prototype

4 • Bilinear map

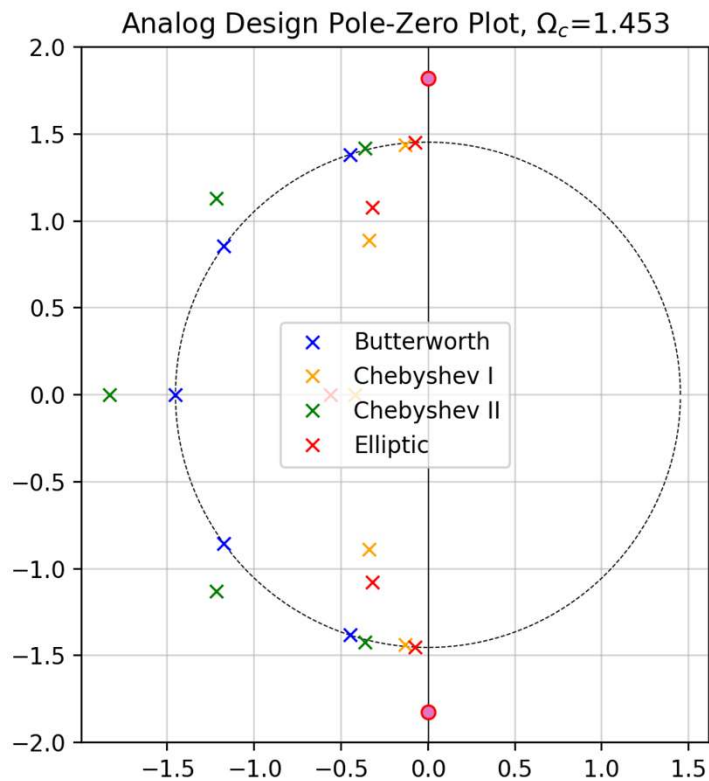
substitute $s(z)$ to obtain the digital filter

Prewarping Digital Specification and Using Bilinear Transform to Convert Analog Prototype to Digital IIR

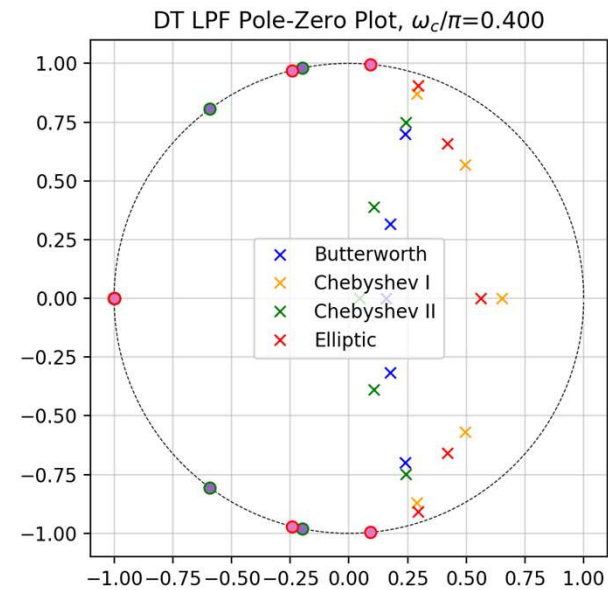
$$\Omega_c = \frac{2}{T} \tan\left(\frac{\omega_c}{2}\right) = 1.453$$

Prewarp
 $\omega_c, T \rightarrow \Omega_c$

$$\omega_c = 0.4\pi, T = 1$$



Bilinear
 Transform
 s to z plane



WORKED EXAMPLE

Prewarp a desired cutoff at 0.4π rad/sample

$$\omega_c = 0.4\pi, \quad T = 1$$

$$\Omega_c = 2 \tan(0.2\pi) \approx 1.4531 \text{ rad/s}$$

Meaning

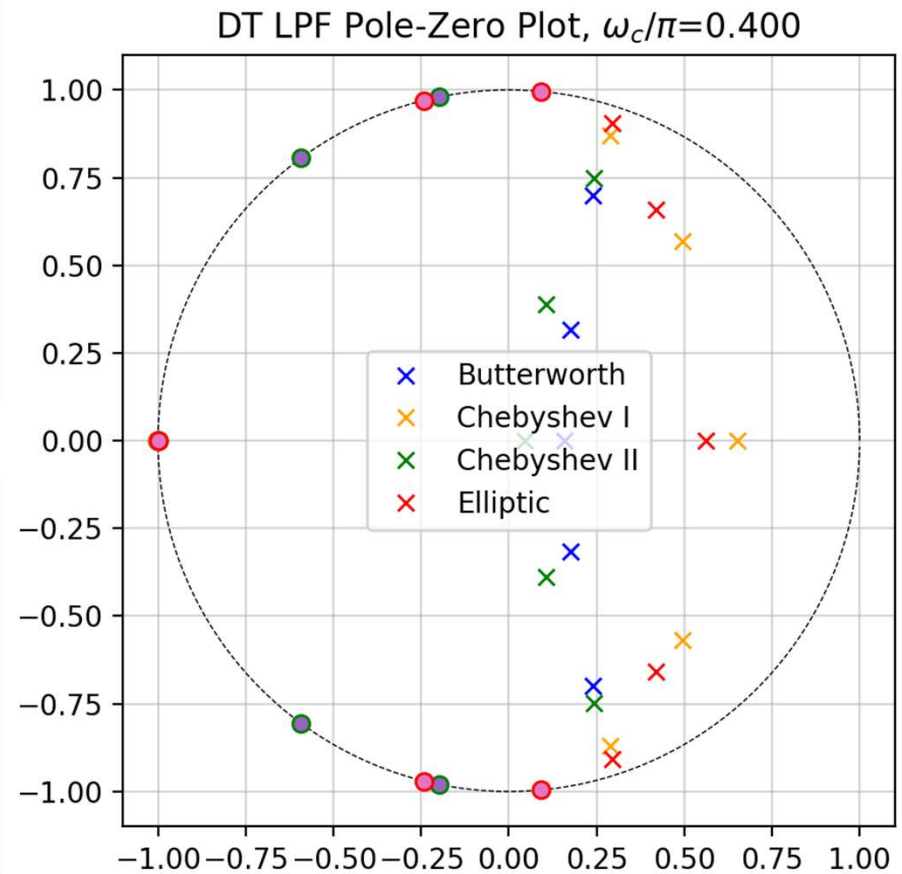
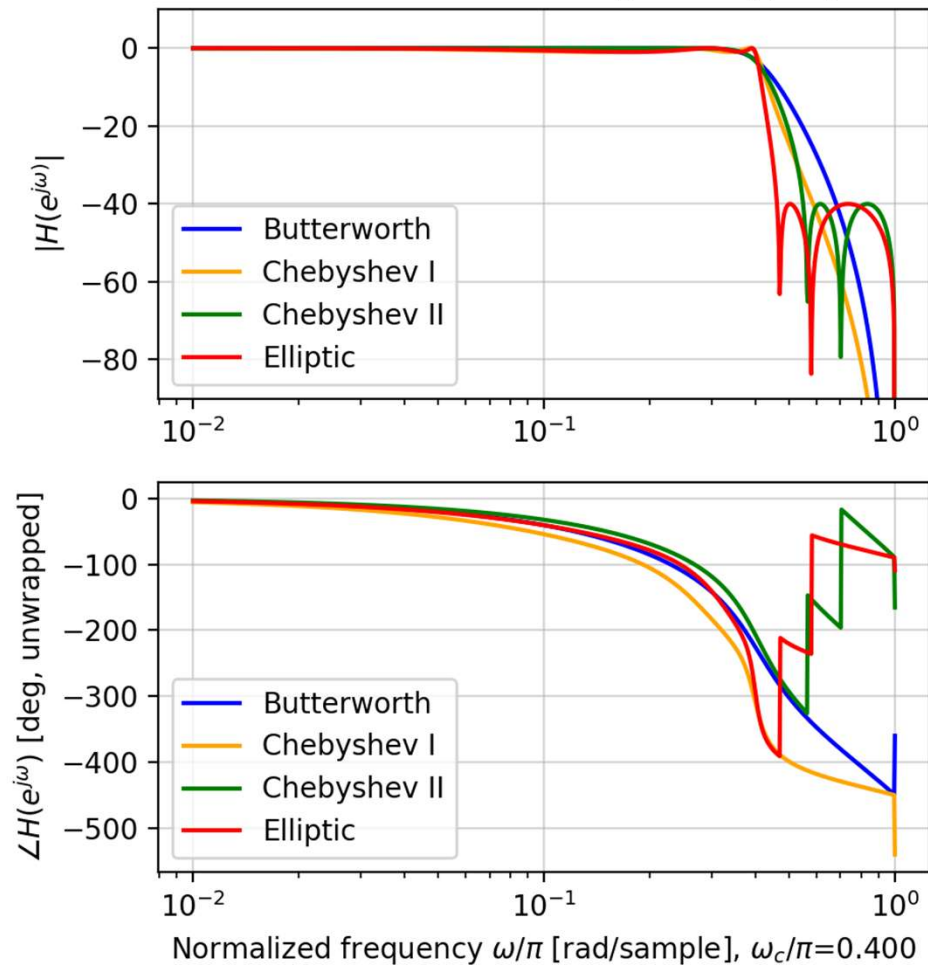
Design the analog prototype at $\Omega_c \approx 1.4531$ —not at $0.4\pi \approx 1.2566$.

Check

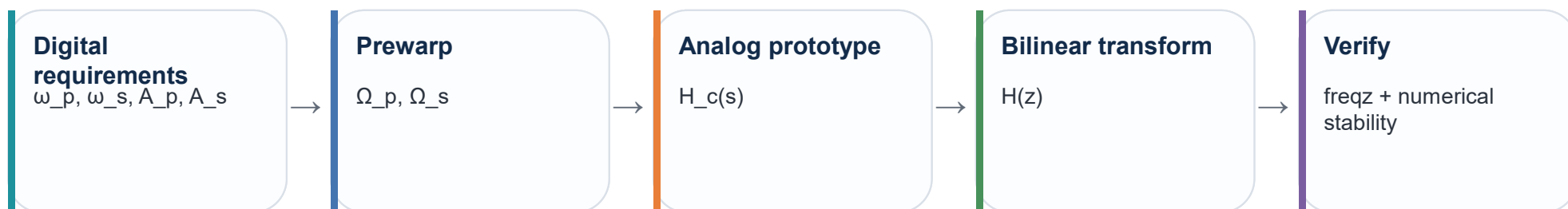
After the bilinear transform, the measured -3 dB point lands at approximately 0.4π .

Verify the Digital Design after Bilinear Transform

DT IIR LPFs from Classical Analog Prototypes, $N=5$, $T=1$



From digital specification to digital IIR



The bilinear transform preserves stability, not exact frequency spacing. Verification is part of the design, not an optional afterthought.

Explicit analog prototype → bilinear transform in SciPy

```
wc = 0.4*np.pi
T = 1.0
Omega_c = (2/T)*np.tan(wc/2)

b_a, a_a = signal.butter(4, Omega_c, analog=True)
b_z, a_z = signal.bilinear(b_a, a_a, fs=1/T)
w, H = signal.freqz(b_z, a_z)
```

Why show the explicit path?

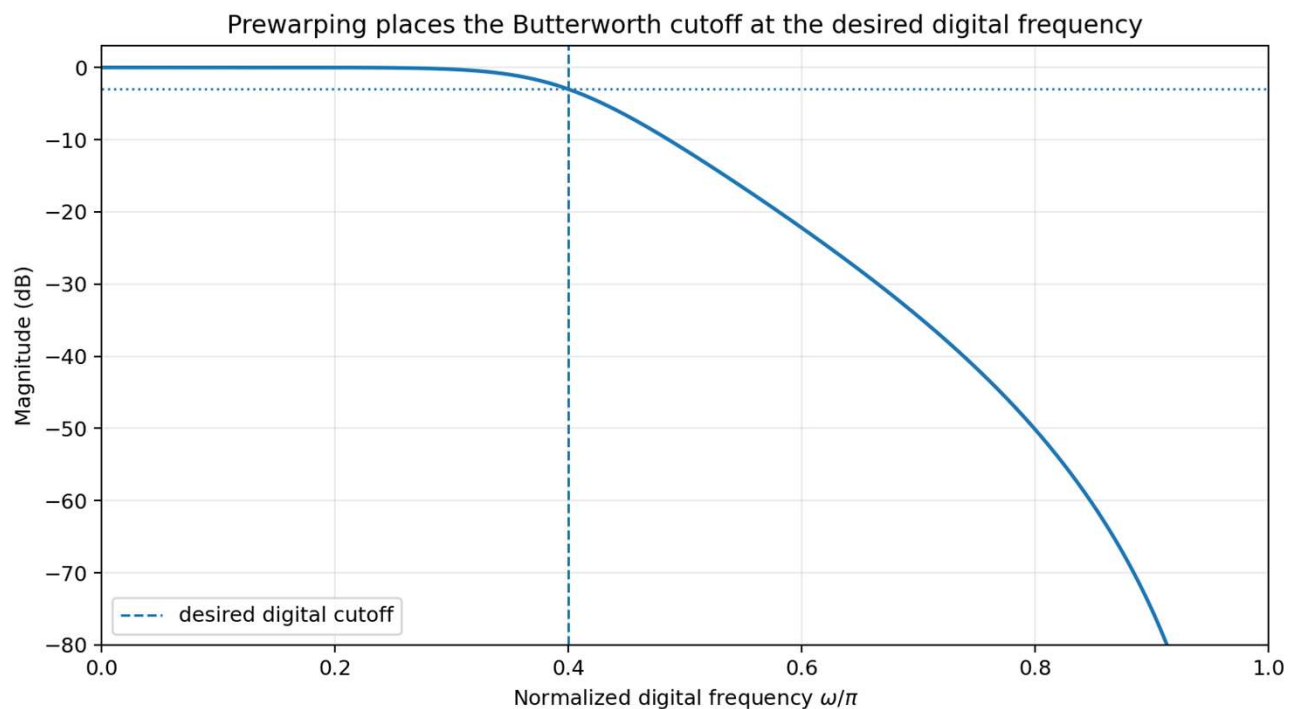
It makes prewarping and the analog prototype visible instead of hiding the mathematics inside one convenience function.

Implementation practice

For deployment, prefer numerically robust second-order-section forms when order becomes substantial.

VERIFICATION

The prewarped design hits the intended digital cutoff



Target

Desired digital cutoff: $\omega_c = 0.4\pi$.

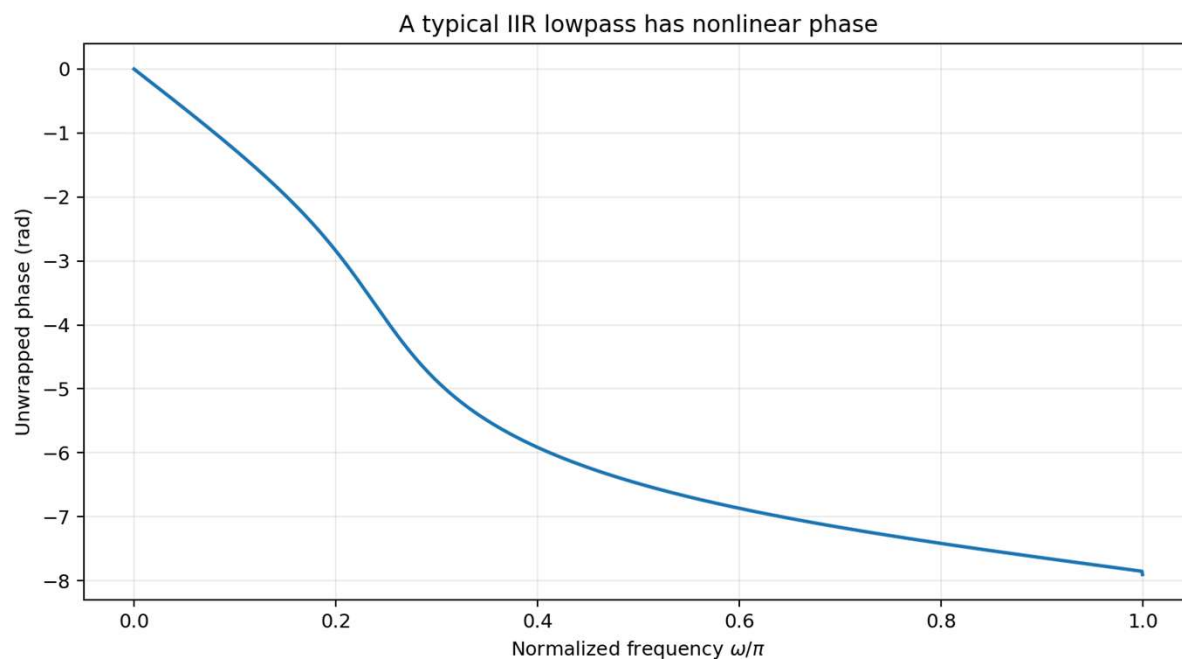
Measured

The script finds the -3 dB point at $\approx 0.3999\pi$.

Good habit

Always verify the digital response after mapping.

Efficient magnitude selectivity does not usually preserve waveform shape



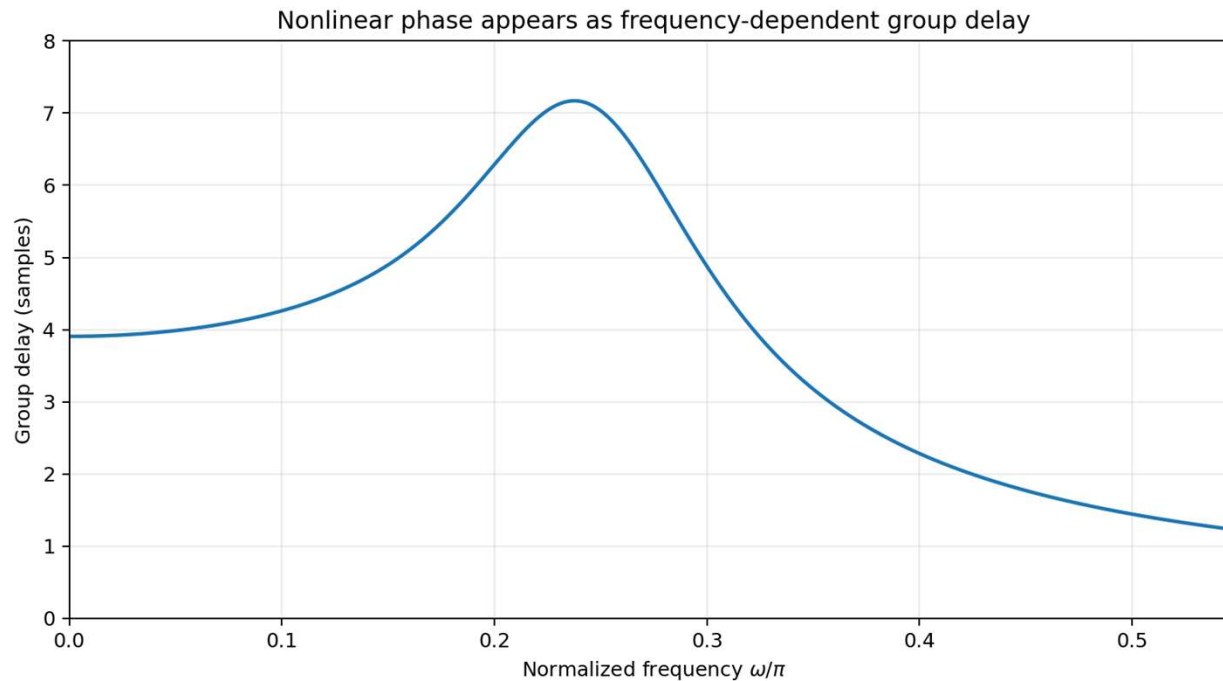
Nonlinear phase

The phase is not a straight line in ω , so group delay varies with frequency.

ML implication

If timing relationships are part of the feature, causal IIR preprocessing can alter them. That may or may not matter depending on the task.

Frequency-dependent group delay makes the distortion easier to see



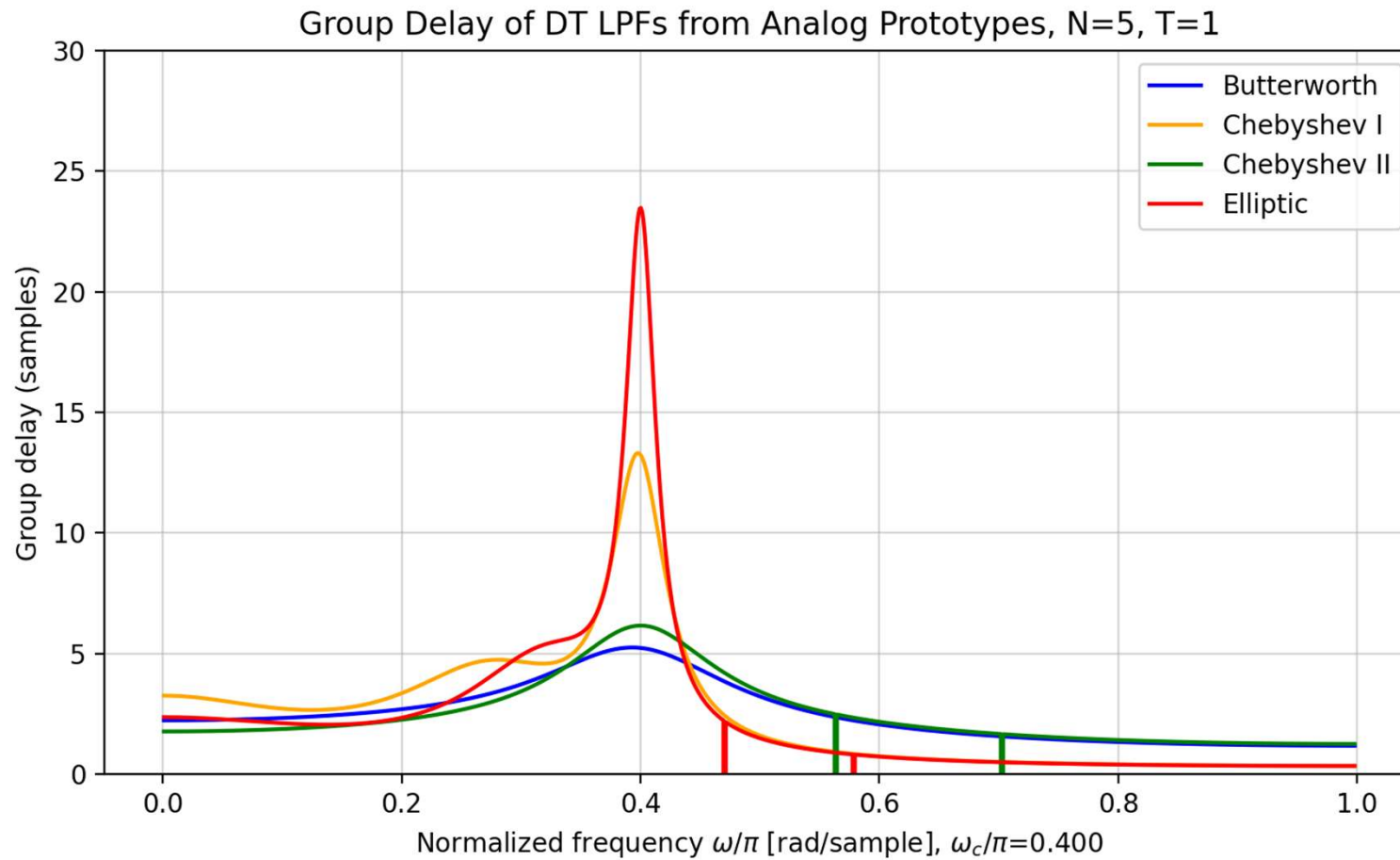
Contrast with Lecture 7

The symmetric FIR had nearly constant τ_g . Here the delay changes across the passband/transition.

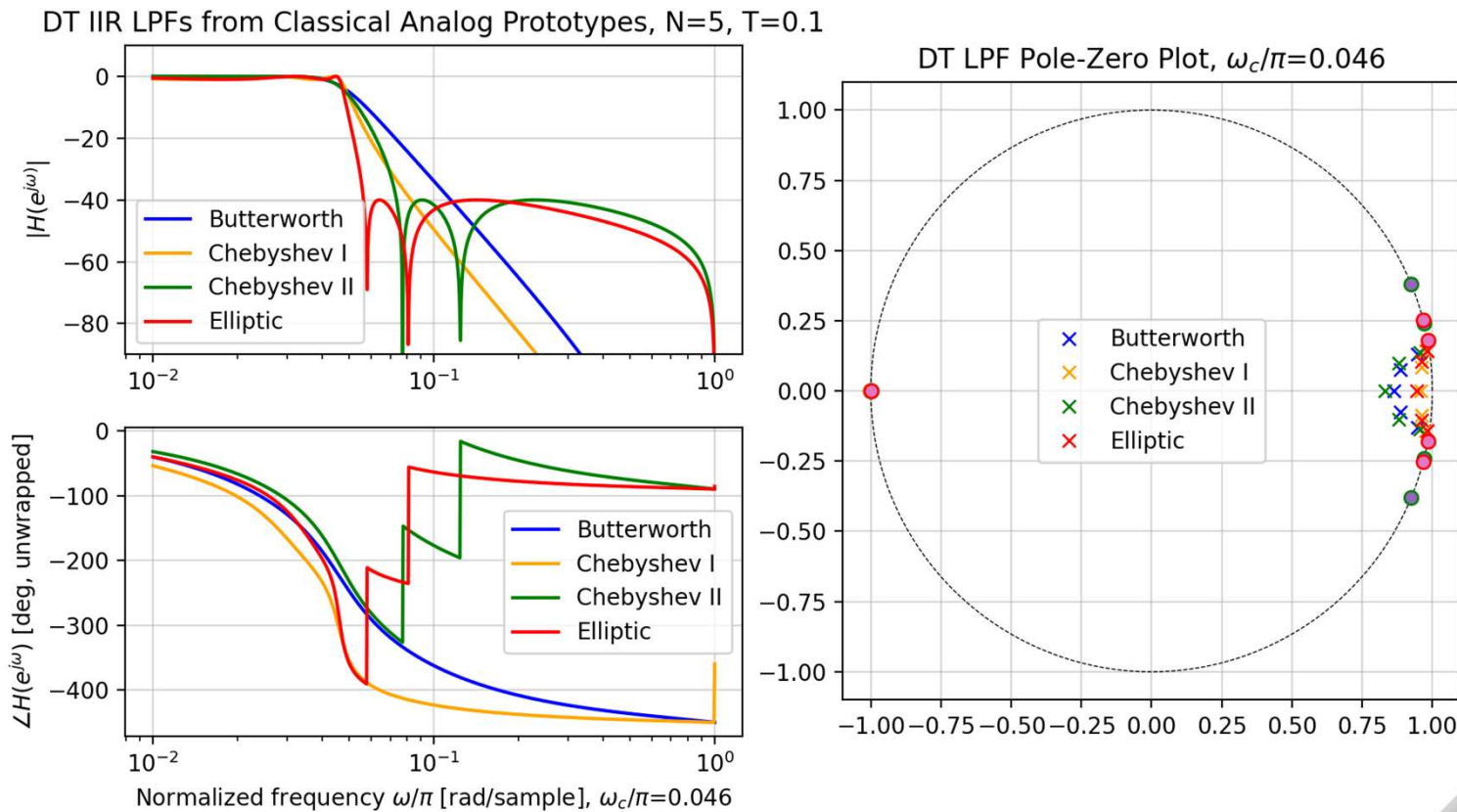
Practical question

Does the downstream model care about absolute timing, relative timing, or only long-term spectral statistics?

Group Delay for the Different IIR Filter Types

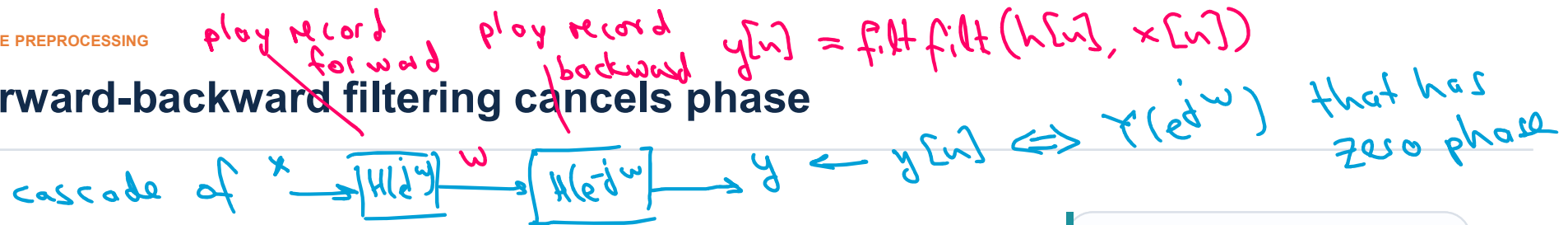


What Happens if $T \rightarrow T/10$ or $f_s \rightarrow 10f_s$?



Poles congregate near $z = 1$. Finite numerical precision can lead to instability.

Forward-backward filtering cancels phase



forward: $H(e^{j\omega})$, backward: $H(e^{-j\omega})$

Net phase

For real-coefficient filters, the phases cancel: zero-phase distortion.

$$H_{fb}(e^{j\omega}) = H(e^{j\omega})H(e^{-j\omega}) = |H(e^{j\omega})|^2$$

real valued coefficients in $H(e^{j\omega})$

$$H(e^{j\omega}) = \sum_n h[n] e^{-j\omega n}$$

$$H^*(e^{j\omega}) = \sum_n \underbrace{h^*[n]}_{=h[n] \text{ if real valued}} e^{j\omega n} = \sum_n h[n] e^{j\omega n} = H(e^{-j\omega})$$

$$\Rightarrow |H(e^{j\omega})|^2 = H(e^{j\omega}) H^*(e^{j\omega}) = H(e^{j\omega}) H(e^{-j\omega})$$

$h[n]$ forward $h[n]$ backward

But magnitude changes

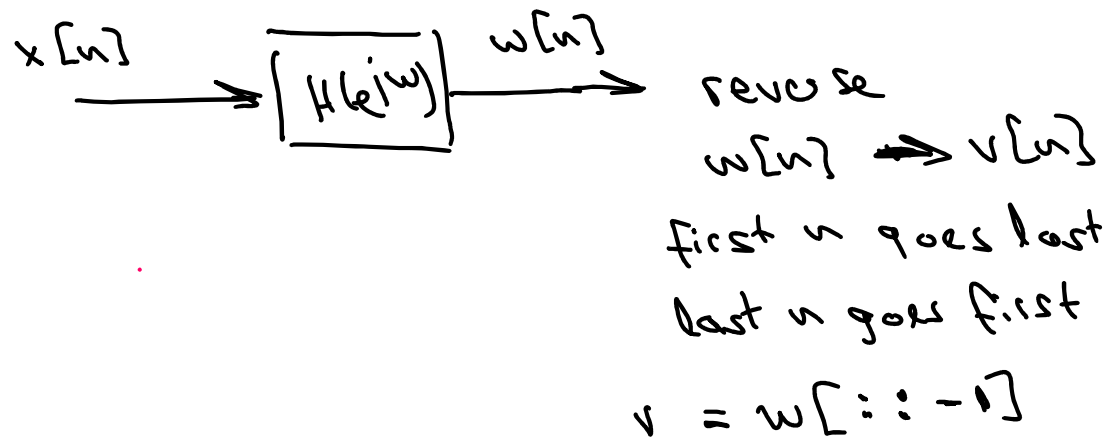
The magnitude is squared, so attenuation in dB is approximately doubled.

And causality is gone

The backward pass uses future samples relative to a streaming implementation.

$$|H(e^{j\omega})|^2 = H(e^{j\omega}) \cdot \underline{H(e^{-j\omega})}$$

This is what
scipy.signal.filtfilt() does



$$y[n] = v[n] * h[n]$$

$$Y(e^{j\omega}) = V(e^{j\omega}) \cdot H(e^{j\omega})$$

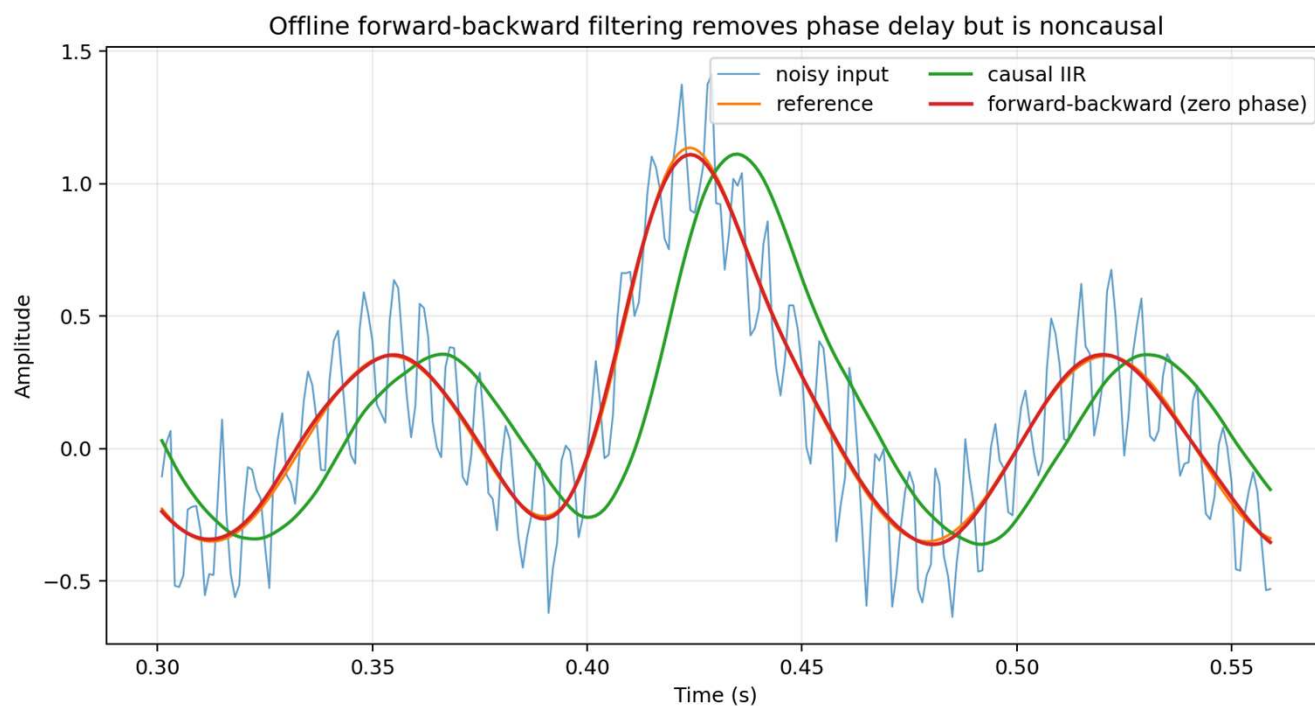
$$= W(e^{j\omega}) H(e^{-j\omega})$$

Good thing: phase of output $y[n]$ is zero

To consider: magnitude of $H(e^{j\omega})$ filter has been applied twice

Cannot be done in real-time

Compare causal IIR delay with forward-backward zero phase



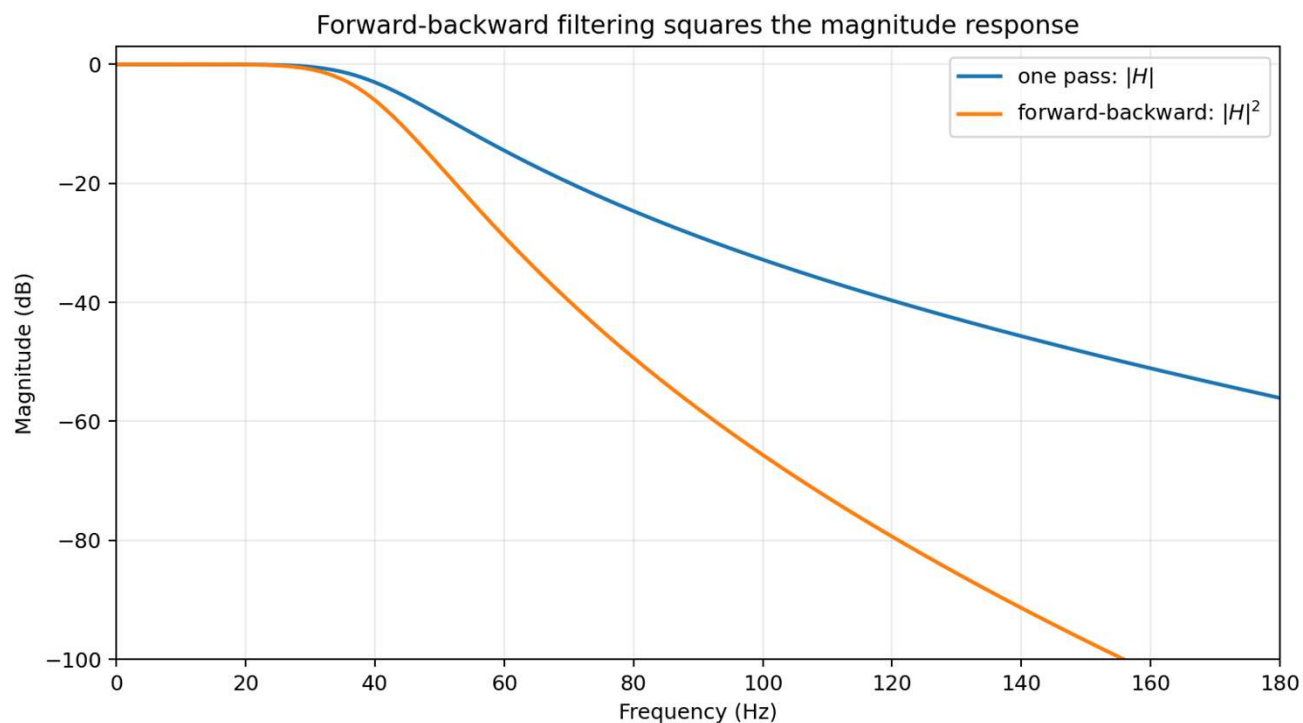
Causal filter

The transient is delayed and reshaped according to the filter phase/group delay.

Forward-backward

The transient aligns much more closely in time—but the method is offline and noncausal.

Zero phase does not mean “the same filter twice for free”



One pass

Magnitude is $|H|$.

Two passes

Magnitude becomes $|H|^2$; order/effective selectivity are increased.

Design implication

If you need a final -3 dB edge after forward-backward filtering, design with the two-pass response in mind.

CAVEATS

Zero-phase preprocessing can quietly change the learning problem

Noncausal

Future samples are used; not reproducible in a strict real-time stream.

Boundary transients

Finite-record initial conditions and padding can distort the beginning/end.

Magnitude squared

The effective response is not the same as one causal pass.

Train/serve skew

A model trained with future-aware filtering may see a different distribution when deployed causally.

Good engineering question: “Will the exact same preprocessing be available when the model is actually used?”

Where a classical IIR can fit



Appropriate example

A known high-frequency interference band is irrelevant to the classification task and can be attenuated efficiently before features are computed.

Question before filtering

Could the “noise” contain class information? Fixed preprocessing is an inductive assumption and should be validated.

FIR, IIR, or learned front end?

Choice	Strength	Cost / risk	Good fit
Linear-phase FIR	controlled phase + finite support	more taps / delay	waveform alignment matters
IIR	sharp response with few coefficients	nonlinear phase; feedback	efficient causal preprocessing
filtfilt IIR	zero phase offline	noncausal; $ H ^2$; edges	offline dataset preparation
Learned Conv1d	task-adaptive front end	needs data/validation	end-to-end models

Lecture 8 synthesis

Prototype

Choose where ripple is allowed and how much transition width you need.

Bilinear map

Stable analog LHP $\rightarrow$ stable digital unit disk.

Prewarp

Critical frequencies must be mapped through $\tan(\omega/2)$.

Preprocessing

Phase, causality, edges, and deployment consistency are part of the ML problem.

Minute paper: Why can a zero-phase offline preprocessing pipeline give misleading confidence about a streaming deployment?